Reverse-check review package — `Sextets_gen`

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	`Sextets/model/Sextets_gen.fr`
original model name	`Sextets_gen` (hidden from the agent)
paper	Sextets/text/0909.2666.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LSextetKin (:=)

```
DC[six1bar[k], mu] DC[six1[k], mu] - MSIX1^2 six1bar[k] six1[k] + DC[six2bar[k], mu]
 ↪ DC[six2[k], mu] - MSIX2^2 six2bar[k] six2[k] + DC[six3bar[k], mu] DC[six3[k], mu] -
 ↪ MSIX3^2 six3bar[k] six3[k]
```

LD11 (:=)

```
2 Sqrt[2] (K6bar[k, i, j] six1[k] LQQR[n, m] ProjP[s, r] dqbar[s, n, i].CC[uq][r, m, j] +
 ↪ K6bar[k, i, j] six1[k] LUDL[n, m] ProjM[s, r] dqbar[s, n, i].CC[uq][r, m, j])
```

LD1 (:=)

```
LD11 + HC[LD11]
```

LD21 (:=)

```
2 Sqrt[2] K6bar[k, i, j] six2[k] LDDL[n, m] ProjM[s, r] dqbar[s, n, i].CC[dq][r, m, j]
```

LD2 (:=)

```
LD21 + HC[LD21]
```

LD31 (:=)

```
2 Sqrt[2] K6bar[k, i, j] six3[k] LUUL[n, m] ProjM[s, r] uqbar[s, n, i].CC[uq][r, m, j]
```

LD3 (:=)

```
LD31 + HC[LD31]
```

LD (:=)

```
LD1 + LD2 + LD3
```

LPot (:=)

```
ExpandIndices[LHS1 Phibar[ii] Phi[ii] six1bar[k] six1[k] + LHS2 Phibar[ii] Phi[ii]
 ↪  six2bar[k] six2[k] + LHS3 Phibar[ii] Phi[ii] six3bar[k] six3[k] + LSS11 six1bar[k1]
 ↪  six1[k1] six1bar[k2] six1[k2] + LSS121 six1bar[k1] six1[k1] six2bar[k2] six2[k2] + LSS122
 ↪  six1bar[k1] six1[k2] six2bar[k2] six2[k1] + LSS131 six1bar[k1] six1[k1] six3bar[k2]
 ↪  six3[k2] + LSS132 six1bar[k1] six1[k2] six3bar[k2] six3[k1] + LSS22 six2bar[k1] six2[k1]
 ↪  six2bar[k2] six2[k2] + LSS231 six2bar[k1] six2[k1] six3bar[k2] six3[k2] + LSS232
 ↪  six2bar[k1] six2[k2] six3bar[k2] six3[k1] + LSS33 six3bar[k1] six3[k1] six3bar[k2]
 ↪  six3[k2], FlavorExpand -> {SU2W, SU2D}]
```

LSextet (:=)

```
LSextetKin + LD1 + LD2 + LD3 + LPot
```

Blank-slate reconstruction

Reconstructed Physics Content of `sanitized.fr`

Conventions

Generation indices are $n, m = 1, 2, 3$. Fundamental color indices are $i, j = 1, 2, 3$. Sextet color indices are $k, k_1, k_2 = 1, \ldots, 6$. The tensor

$$K^*_{kij} \equiv \texttt{K6bar[k,i,j]}$$

is the color Clebsch tensor contracting a color-sextet scalar with two color-triplet quarks; it is symmetric in the two fundamental color indices.

The three new scalar fields are denoted

$$S_1^k \equiv \texttt{six1[k]}, \qquad S_2^k \equiv \texttt{six2[k]}, \qquad S_3^k \equiv \texttt{six3[k]}.$$

FeynRules projectors are translated as

$$\texttt{ProjP} = P_R = \frac{1+\gamma^5}{2}, \qquad \texttt{ProjM} = P_L = \frac{1-\gamma^5}{2}.$$

The charge-conjugated spinor is

$$q^c \equiv C\bar{q}^T.$$

The scalar fields are $SU(2)_L$ singlets. Their covariant derivative is therefore

$$D_\mu S_a = \left[\partial_\mu + i g_s G^A_\mu (T^A_6) + i g' Y_a B_\mu\right] S_a,$$

or, after electroweak rotation,

$$D_\mu S_a = \left[\partial_\mu + i g_s G^A_\mu (T^A_6) + i e Q_a A_\mu - i e Q_a \tan\theta_W Z_\mu\right] S_a,$$

with $Q_a = Y_a$ for these $SU(2)_L$-singlet scalars.

Lagrangian

LSextetKin

$$\mathcal{L}_{\texttt{LSextetKin}} \supset (D_\mu S_1)^\dagger_k (D^\mu S_1)^k - M^2_{\texttt{SIX1}}\, S^\dagger_{1k} S^k_1.$$

$$\mathcal{L}_{\texttt{LSextetKin}} \supset (D_\mu S_2)^\dagger_k (D^\mu S_2)^k - M^2_{\texttt{SIX2}}\, S^\dagger_{2k} S^k_2.$$

$$\mathcal{L}_{\texttt{LSextetKin}} \supset (D_\mu S_3)^\dagger_k (D^\mu S_3)^k - M^2_{\texttt{SIX3}}\, S^\dagger_{3k} S^k_3.$$

`LD11`

$$\mathcal{L}_{\texttt{LD11}} = 2\sqrt{2}\, K^*_{kij}\, S_1^k \left[\lambda^{nm}_{\texttt{QQ}}\, \bar{d}_{ni} P_R u^c_{mj} + \lambda^{nm}_{\texttt{UD}}\, \bar{d}_{ni} P_L u^c_{mj}\right],$$

where

$$\lambda^{nm}_{\texttt{QQ}} \equiv \texttt{LQQR[n,m]}, \qquad \lambda^{nm}_{\texttt{UD}} \equiv \texttt{LUDL[n,m]}.$$

`LD1`

$$\mathcal{L}_{\texttt{LD1}} = \mathcal{L}_{\texttt{LD11}} + \mathcal{L}^\dagger_{\texttt{LD11}}.$$

`LD21`

$$\mathcal{L}_{\texttt{LD21}} = 2\sqrt{2}\, K^*_{kij}\, S_2^k\, \lambda^{nm}_{\texttt{DD}}\, \bar{d}_{ni} P_L d^c_{mj},$$

where

$$\lambda^{nm}_{\texttt{DD}} \equiv \texttt{LDDL[n,m]}.$$

`LD2`

$$\mathcal{L}_{\texttt{LD2}} = \mathcal{L}_{\texttt{LD21}} + \mathcal{L}^\dagger_{\texttt{LD21}}.$$

`LD31`

$$\mathcal{L}_{\texttt{LD31}} = 2\sqrt{2}\, K^*_{kij}\, S_3^k\, \lambda^{nm}_{\texttt{UU}}\, \bar{u}_{ni} P_L u^c_{mj},$$

where

$$\lambda^{nm}_{\texttt{UU}} \equiv \texttt{LUUL[n,m]}.$$

`LD3`

$$\mathcal{L}_{\texttt{LD3}} = \mathcal{L}_{\texttt{LD31}} + \mathcal{L}^\dagger_{\texttt{LD31}}.$$

`LD`

$$\mathcal{L}_{\texttt{LD}} = \mathcal{L}_{\texttt{LD1}} + \mathcal{L}_{\texttt{LD2}} + \mathcal{L}_{\texttt{LD3}}.$$

`LPot`

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{HS1}}\, (\Phi_i^\dagger \Phi_i)\, S^\dagger_{1k} S_1^k.$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{HS2}}\, (\Phi_i^\dagger \Phi_i)\, S^\dagger_{2k} S_2^k.$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{HS3}}\, (\Phi_i^\dagger \Phi_i)\, S^\dagger_{3k} S_3^k.$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{SS11}}\, (S^\dagger_{1k_1} S_1^{k_1})(S^\dagger_{1k_2} S_1^{k_2}).$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{SS121}}\, (S^\dagger_{1k_1} S_1^{k_1})(S^\dagger_{2k_2} S_2^{k_2}).$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{SS122}}\, S^\dagger_{1k_1} S_1^{k_2} S^\dagger_{2k_2} S_2^{k_1}.$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{SS131}}\, (S^\dagger_{1k_1} S_1^{k_1})(S^\dagger_{3k_2} S_3^{k_2}).$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{SS132}}\, S^{\dagger}_{1k_1} S_1^{k_2} S^{\dagger}_{3k_2} S_3^{k_1}.$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{SS22}}\, (S^{\dagger}_{2k_1} S_2^{k_1})(S^{\dagger}_{2k_2} S_2^{k_2}).$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{SS231}}\, (S^{\dagger}_{2k_1} S_2^{k_1})(S^{\dagger}_{3k_2} S_3^{k_2}).$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{SS232}}\, S^{\dagger}_{2k_1} S_2^{k_2} S^{\dagger}_{3k_2} S_3^{k_1}.$$

$$\mathcal{L}_{\texttt{LPot}} \supset \lambda_{\texttt{SS33}}\, (S^{\dagger}_{3k_1} S_3^{k_1})(S^{\dagger}_{3k_2} S_3^{k_2}).$$

`LSextet`

$$\mathcal{L}_{\texttt{LSextet}} = \mathcal{L}_{\texttt{LSextetKin}} + \mathcal{L}_{\texttt{LD1}} + \mathcal{L}_{\texttt{LD2}} + \mathcal{L}_{\texttt{LD3}} + \mathcal{L}_{\texttt{LPot}}.$$

Field Table

`.fr` class	Field symbol	Spin	$SU(3)_C$ rep	$SU(2)_L$ rep	$U(1)$ charge / hyper-charge	Self-conjugate?	Mass
`six1`	S_1	0	**6**	**1**	$Q = Y = 1/3$	No	`MSIX1 = 500`
`six2`	S_2	0	**6**	**1**	$Q = Y = -2/3$	No	`MSIX2 = 500`
`six3`	S_3	0	**6**	**1**	$Q = Y = 4/3$	No	`MSIX3 = 500`

Parameters

External parameter	Value in file	Multiplies	Physical meaning
`LQQRR[n,m]`	diagonal 0.1, off-diagonal 0	real part of `LQQR[n,m]` in $S_1\, \bar{d}_n P_R u^c_m$	real part of complex diquark Yukawa coupling
`LQQRI[n,m]`	0	imaginary part of `LQQR[n,m]` in $S_1\, \bar{d}_n P_R u^c_m$	imaginary part / CP phase of diquark coupling
`LUDLR[n,m]`	diagonal 0.1, off-diagonal 0	real part of `LUDL[n,m]` in $S_1\, \bar{d}_n P_L u^c_m$	real part of complex diquark Yukawa coupling
`LUDLI[n,m]`	0	imaginary part of `LUDL[n,m]` in $S_1\, \bar{d}_n P_L u^c_m$	imaginary part / CP phase of diquark coupling
`LUULR[n,m]`	diagonal 0.1, off-diagonal 0	real part of `LUUL[n,m]` in $S_3\, \bar{u}_n P_L u^c_m$	real part of complex up-type diquark Yukawa coupling

External parameter	Value in file	Multiplies	Physical meaning
`LUULI[n,m]`	0	imaginary part of `LUUL[n,m]` in $S_3\,\bar{u}_n P_L u^c_m$	imaginary part / CP phase of up-type diquark coupling
`LDDLR[n,m]`	diagonal 0.1, off-diagonal 0	real part of `LDDL[n,m]` in $S_2\,\bar{d}_n P_L d^c_m$	real part of complex down-type diquark Yukawa coupling
`LDDLI[n,m]`	0	imaginary part of `LDDL[n,m]` in $S_2\,\bar{d}_n P_L d^c_m$	imaginary part / CP phase of down-type diquark coupling
`LHS1`	0.1	$(\Phi^\dagger\Phi)(S_1^\dagger S_1)$	Higgs-portal quartic coupling
`LHS2`	0.1	$(\Phi^\dagger\Phi)(S_2^\dagger S_2)$	Higgs-portal quartic coupling
`LHS3`	0.1	$(\Phi^\dagger\Phi)(S_3^\dagger S_3)$	Higgs-portal quartic coupling
`LSS11`	0.1	$(S_1^\dagger S_1)^2$	scalar self-quartic coupling
`LSS121`	0.1	$(S_1^\dagger S_1)(S_2^\dagger S_2)$	mixed scalar quartic coupling
`LSS122`	0.1	$S^\dagger_{1\,k_1} S_1^{k_2} S^\dagger_{2\,k_2} S_2^{k_1}$	independent color-contracted mixed scalar quartic
`LSS131`	0.1	$(S_1^\dagger S_1)(S_3^\dagger S_3)$	mixed scalar quartic coupling
`LSS132`	0.1	$S^\dagger_{1\,k_1} S_1^{k_2} S^\dagger_{3\,k_2} S_3^{k_1}$	independent color-contracted mixed scalar quartic
`LSS22`	0.1	$(S_2^\dagger S_2)^2$	scalar self-quartic coupling
`LSS231`	0.1	$(S_2^\dagger S_2)(S_3^\dagger S_3)$	mixed scalar quartic coupling
`LSS232`	0.1	$S^\dagger_{2\,k_1} S_2^{k_2} S^\dagger_{3\,k_2} S_3^{k_1}$	independent color-contracted mixed scalar quartic
`LSS33`	0.1	$(S_3^\dagger S_3)^2$	scalar self-quartic coupling

The internal complex couplings are

$$\mathtt{LQQR}_{nm} = \mathtt{LQQRR}_{nm} + i\,\mathtt{LQQRI}_{nm},$$

$$\mathtt{LUDL}_{nm} = \mathtt{LUDLR}_{nm} + i\,\mathtt{LUDLI}_{nm},$$

$$\mathtt{LUUL}_{nm} = \mathtt{LUULR}_{nm} + i\,\mathtt{LUULI}_{nm},$$

$$\mathtt{LDDL}_{nm} = \mathtt{LDDLR}_{nm} + i\,\mathtt{LDDLI}_{nm}.$$

Physics Summary

The file encodes three complex scalar color-sextet, electroweak-singlet fields with electric charges 1/3, −2/3, and 4/3. They have QCD and hypercharge gauge interactions, Higgs-portal and scalar self-quartic interactions, and diquark Yukawa-type couplings to pairs of quarks through charge-conjugated spinors with explicitly chiral projectors.

The model mediates quark-quark resonant or virtual processes such as ud, dd, and uu scattering through scalar color-sextet exchange, with generation structure controlled by the complex matrices `LQQR`, `LUDL`, `LDDL`, and `LUUL`.

Paper cross-check

Comparison of `reconstruction.md` Against the Paper Model

Paper Model Location

The paper defines the model in **Section 2, "Model And Constraints"**. The relevant ingredients are:

- **Table 1**: possible electroweak quantum numbers and allowed quark bilinears for scalar diquarks. Each listed state may be either an $SU(3)_C$ antitriplet $\bar{\mathbf{3}}$ or sextet $\mathbf{6}$.
- **Eq. (2.1)**: the post-EWSB scalar diquark interaction

$$\mathcal{L} = 2\sqrt{2}\left[\bar{K}^i_{ab} D_i \bar{q}^a (\lambda_L P_L + \lambda_R P_R) q^C_b + \text{h.c.}\right].$$

- **Eq. (2.2)**: example constraints for a sextet diquark coupled to right-handed up-type quarks.
- **Section 3, Eqs. (3.1)-(3.3)**: production process, Born amplitude, and normalization conventions.
- **Appendix A, especially Eqs. (A.6)-(A.16)**: Clebsch-Gordan tensors and sextet/antitriplet color-generator conventions.

Term-By-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Generic scalar diquark D_i, color representation either $\mathbf{6}$ or $\bar{\mathbf{3}}$ (Section 2; Table 1; Eq. (2.1))	Three scalar fields S_1, S_2, S_3, all in color $\mathbf{6}$	missing-in-reconstruction	Reconstruction covers only the sextet branch. The paper explicitly allows both sextet and antitriplet color representations.
Antitriplet color option with antisymmetric flavor couplings (Section 2 after Eq. (2.1); Appendix A, Eq. (A.7))	No antitriplet scalar, no antisymmetric flavor condition	missing-in-reconstruction	The paper states that in the antitriplet case the couplings must be antisymmetric in flavor. Reconstruction is sextet-only.
Sextet Clebsch tensor K^i_{ab}, symmetric in the two triplet indices (Appendix A, Eq. (A.6), completeness Eq. (A.10))	$K^*_{kij} =$ `K6bar[k,i,j]`, symmetric color-sextet contraction	agree	This matches the sextet color structure used by the paper.
Diquark may be scalar; paper concentrates on scalar case (Section 2)	S_1, S_2, S_3 are spin-0 scalars	agree	Reconstruction matches the scalar specialization.
Table 1 singlet state: $SU(2)_L = \mathbf{1}$, $Y = 1/3$, $\lvert Q\rvert = 1/3$, couplings QQ, UD	$S_1 \sim (\mathbf{6}, \mathbf{1}, Y = 1/3)$, couplings to du through `LQQR` and `LUDL`	agree	The two S_1 chiral structures correspond to the paper's QQ and UD entries after translating projectors acting on charge-conjugated spinors.

paper term (eq. ref)	reconstruction term	verdict	notes
Table 1 triplet state: $SU(2)_L = \mathbf{3}$, $Y = 1/3$, $\|Q\| = 1/3, 2/3, 4/3$, coupling QQ	No $SU(2)_L$ triplet diquark multiplet	missing-in-reconstruction	The paper lists this as a possible electroweak assignment, though the later QCD calculation is model-independent after EWSB.
Table 1 singlet state: $SU(2)_L = \mathbf{1}$, $Y = 2/3$, $\|Q\| = 2/3$, coupling DD	$S_2 \sim (\mathbf{6}, \mathbf{1}, Y = -2/3)$, coupling $S_2\, \bar{d} P_L d^C$	disagree	The reconstruction's Yukawa term is gauge invariant for S_2 charge $-2/3$, while Table 1 lists $Y = +2/3$. This may be a D versus $D^\dagger$ naming convention, but the hypercharge sign differs as written.
Table 1 singlet state: $SU(2)_L = \mathbf{1}$, $Y = 4/3$, $\|Q\| = 4/3$, coupling UU	$S_3 \sim (\mathbf{6}, \mathbf{1}, Y = 4/3)$, coupling $S_3\, \bar{u} P_L u^C$	agree	Charge, singlet electroweak representation, and up-type bilinear match the paper's UU entry.
Eq. (2.1): universal coefficient $2\sqrt{2}$ multiplying $\bar{K} D \bar{q}(\lambda_L P_L + \lambda_R P_R) q^C + \text{h.c.}$	All reconstructed diquark Yukawas use $2\sqrt{2}\, K^* S\, \bar{q} P_{L/R} q^C$, with hermitian conjugates in `LD1`, `LD2`, `LD3`	agree	Coefficient and h.c. structure match for the sextet singlet interactions.
Eq. (2.1): generic independent flavor couplings $\lambda_{L,R}$, depending on flavor channel	Complex matrices `LQQR`, `LUDL`, `LDDL`, `LUUL`	agree	Reconstruction's independent coupling matrices are compatible with the paper's model-independent parameterization.
Eq. (2.1): generic chiral projectors P_L, P_R	`ProjM=P_L`, `ProjP=P_R`; terms use P_R for QQ, P_L for UD, DD, UU in the charge-conjugated-spinor notation	agree	With four-component charge-conjugation identities, these projector placements are consistent with the corresponding left-left or right-right quark bilinears.
Eq. (2.2): example sextet UU-type constraints $\lambda_R^{uu}, \lambda_R^{uc} \lesssim 0.1$, $\lambda_R^{cc} \simeq 0$	Defaults all listed Yukawa matrices diagonal 0.1, off-diagonal 0	disagree	The paper gives illustrative constraints, not a universal default coupling matrix. In particular, it singles out right-handed up-type sextet couplings, not all QQ, UD, DD, UU channels.
Section 3, Eq. (3.2): Born amplitude $qq \to D$ with $K_{iab}(\lambda_L P_L + \lambda_R P_R) C^\dagger$	No explicit amplitude, but Yukawa Lagrangian implies the same sextet vertex up to standard Feynman-rule signs	agree	The missing overall $-i$ is not a Lagrangian-content mismatch.

paper term (eq. ref)	reconstruction term	verdict	notes
Paper treats a single generic diquark mass m_D in production formulas (Eqs. (3.3)-(3.4))	Three independent masses `MSIX1`, `MSIX2`, `MSIX3`, all defaulting to 500	extra-in-reconstruction	Multiple specific mass parameters are implementation details beyond the paper's generic m_D.
Paper says colored states interact with gluons; Appendix A defines generators in the diquark representation (Section 2; Appendix A.3)	Sextet covariant derivatives with $g_s G^A_\mu T^A_6$	agree	Reconstruction implements the QCD gauge interactions implied by the paper for the sextet case.
Paper does not specify electroweak gauge interactions after saying it is "not concerned about the electroweak structure" (Section 2)	Full $U(1)_Y$, photon, and Z couplings for S_1, S_2, S_3	extra-in-reconstruction	These follow from chosen Table 1 electroweak assignments but are not part of the paper's operative Lagrangian.
Paper gives no explicit scalar kinetic or mass Lagrangian	`LSextetKin` with canonical kinetic and mass terms for all three sextets	extra-in-reconstruction	Canonical kinetic terms are physically expected, but they are not written as part of the paper's defined interaction Lagrangian.
Paper gives no Higgs-portal interactions	`LHS1`, `LHS2`, `LHS3`: $(\Phi^\dagger\Phi)(S_a^\dagger S_a)$	extra-in-reconstruction	No such terms appear in the paper.
Paper gives no scalar self-quartic potential	`LSS11`, `LSS22`, `LSS33` and mixed `LSS12*`, `LSS13*`, `LSS23*` quartics	extra-in-reconstruction	These are implementation-level additions, not part of the phenomenological model used in the paper.
Paper ignores diquark decay and treats the produced diquark as stable on shell (Section 4)	Reconstruction includes only production-type diquark Yukawas and no explicit decay-sector additions beyond h.c.	agree	The Yukawa h.c. would mediate the inverse process/decay, but the reconstruction does not add a separate decay model.

Disagreements and Human Checks

- **Missing antitriplet branch** — severity: substantive. A human should check whether the implementation was intended to reproduce the full paper parameterization or only the sextet specialization used in parts of the numerical study.
- **Missing $SU(2)_L$ triplet QQ electroweak option** — severity: substantive. A human should check whether Table 1 is meant as required model content for this implementation or only a catalogue of possible diquark assignments.
- **DD hypercharge sign mismatch: paper Table 1 lists $Y = +2/3$, reconstruction uses $Y = -2/3$** — severity: convention if this is only a D versus $D^\dagger$ naming choice, substantive if electroweak gauge vertices are used directly. A human should trace which field, S_2 or $S_2^\dagger$, is intended to be the physical DD-coupled diquark of the paper.
- **Universal default Yukawa values in reconstruction are broader than Eq. (2.2)** — severity: substantive for phenomenology, cosmetic for pure structure. A human should check whether the implementation defaults were chosen merely as placeholders or are being interpreted as paper-derived constraints.

- **Three separate sextet masses instead of one generic m_D** — severity: convention. A human should check whether analyses using the implementation set these masses consistently when comparing to the paper's single-diquark production formulas.
- **Electroweak gauge interactions are reconstructed but not part of the paper's operative Lagrangian** — severity: convention. A human should check whether any calculation uses photon or Z interactions, since the paper's results are QCD-focused and do not validate those vertices.
- **Higgs-portal and sextet scalar quartic potential terms are extra** — severity: substantive if used in predictions beyond single production. A human should check whether these potential terms came from an implementation extension rather than from the paper.

Overall Assessment

The reconstruction captures the paper's central sextet scalar diquark Yukawa structure well: the $2\sqrt{2}$ normalization, charge-conjugated quark bilinears, sextet Clebsch tensor, hermitian conjugates, and the singlet QQ/UD and UU channels are broadly consistent with Section 2 and Appendix A. The main caveat is scope: the paper defines a generic model-independent diquark parameterization, including antitriplet and multiple electroweak possibilities, while the reconstruction describes a specific sextet-only implementation with three singlet fields and a much larger scalar/gauge sector. The most physics-sensitive point to audit is the DD state's hypercharge sign and field-conjugation convention, followed by whether the added scalar potential and default coupling matrices are being treated as paper content or implementation extras.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 27 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: ________________________________ Date: __________________
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: